

A Review of Visual SLAM Based on Unmanned Systems

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Abstract—Simultaneous Localization and Mapping (SLAM) consists of the immediate construction of the environment and the state estimation of the robot in it, while Visual SLAM (VSLAM) is the use of cameras and other visual sensors for SLAM. VSLAM has become an important part of mobile robots, drones, unmanned vehicles and other unmanned systems in unknown environments to achieve full-scale navigation and environmental perception. First, the principle of architecture, the mathematical models, the current research status and the algorithms of each part have been reviewed. Then, the research hotspots and current facing challenges on VSLAM were summarized from three parts: (i) VSLAM and deep learning; (ii) data processing of multi-sensor; (iii) VSLAM in visual/inertial navigation. Moreover, the research trend of VSLAM were further analyzed, including (i) deep learning and deep estimation, (ii) active and multi-robot VSLAM and (iii) semantic VSLAM. At last, the future development of VSLAM was discussed, which may provide a certain guiding significance for researchers in this area.

Keywords—simultaneous localization and mapping; environment perception; state estimation; deep learning

I. INTRODUCTION

Nowadays, the development of unmanned systems, such as micro aerial vehicle (MAV) and automated guided vehicle (AGV), has shown the trend of autonomy, miniaturization and intelligence^[1]. The application scenarios of them have gradually expanded from military fields (material delivery, battlefield reconnaissance and cooperative operation) to civil fields (warehouse management, disaster rescue, urban security, resource exploration and power patrol). Generally, these new scenarios have high dynamic, unknown and closed nature. Relying on their own sensors, the unmanned systems, without environmental prior information and reliable external auxiliary information sources such as Global Positioning System (GPS) and telemetry systems, should have the ability to achieve full-scale navigation, environmental perception and provide other necessary information. Simultaneous Localization and Mapping (SLAM) is the preferred strategy to solve the above-mentioned problems.

To achieve SLAM, unmanned systems require the realization of environmental perception, data processing,

completion of self-positioning and path planning in unfamiliar environments, which has been widely used in the fields of driverless technology, mobile robots, drones and augmented reality. The positioning in SLAM focus on its own understanding, which is the premise of path planning. Therefore, the accurate positioning needs accurate map. Building maps in SLAM focus on external understanding, and the accurate maps coming from accurate positioning. SLAM is the basis of environment perception and intelligent navigation in mobile robots while the positions of robots is the premise of its navigation. Therefore, unmanned systems such as mobile robots need to create environmental maps while calculating their own positions in unfamiliar environments. Three urgent problems are first proposed by Leonard on mobile robot navigation: (i) where I am, (ii) where will I go and (iii) how can I get there^[2]. Up to now, the development of SLAM has gone through three important stages: classical stage (1986-2004), algorithm analysis stage (2004-2015) and robust perception stage (2015-)^[2]. In classical stage, the researches focused on the problems of state estimation in SLAM, including a large number of pose estimation algorithms based on Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF) and Particle Filter (PF). In algorithm analysis stage, researches focused on the observability, convergence, consistency and sparsity in SLAM. In robust perception stage, the researches include the following three aspects: (i) optimizing the adaptability of SLAM systems by adding fault sensing and parameter adaptive modules; (ii) improving intelligent perception (from basic geometric construction to advanced semantic construction); (iii) developing task-driven capability to automatically filter sensor information and deploy sensor resources.

SLAM is essentially a state estimation strategy, and, by sensors, it can be mainly divided into two categories: Laser SLAM (LSLAM) and Visual SLAM (VSLAM). Researches based on LSLAM is relatively mature both in theory and engineering, and it have been employed to solving some industrial works. Image information is the main environment sensing source in VSLAM system, targeting on the calculation of the pose of camera and the construction of three dimensional (3D) map. The process of VSLAM system is generally divided into two stages: (i) inter-frame estimation and (ii) back-end

optimization, which was first proposed and implemented in Parallel Tracking and Mapping (PTAM) system^[3]. PTAM distinguishes the front and back ends and completes the parallelization of feature point tracking and mapping. The real-time response from image to data is required in front-end tracking, while the map optimization is performed at the back-end. Moreover, the PTAM is also the first system to use nonlinear optimization in back-end, in which the key frame mechanism was proposed to optimize its trajectory and map by stringing several key images instead of fine processing each image. For VSLAM system, the inter-frame estimation is based on the sensor information between two adjacent frames to obtain the motion estimation in this time interval, while the back-end optimization refers to the optimization of the path cumulative drift error generated by the previous inter-frame estimation to optimize the historical trajectory after the path closed loop detected by the machine.

A review on VSLAM in unmanned systems is presented in this work, including (i) summary on the development history, structure principle, mathematical models and current research status and algorithms of VSLAM; (ii) discussion on the hotspots and challenges of VSLAM (including VSLAM techniques based on deep learning, multi-sensor deep fusion and visual inertial combined navigation); (iii) discussion on the future development of VSLAM for providing beneficial reference for researchers.

II. VSLAM SYSTEM

A. Architecture of VSLAM

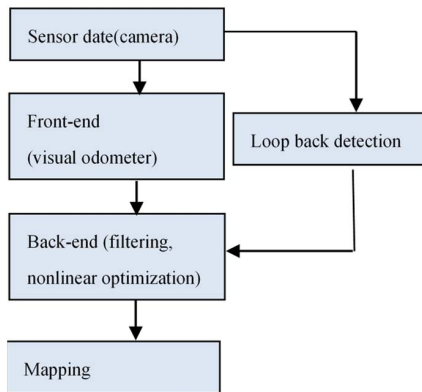


Figure 1. Architecture of VSLAM.

The architecture of the VSLAM system is the same as the classical SLAM framework, which is mainly composed of four modules: front-end (visual odometer, VO), back-end (filtering, graph optimization), loop back detection and Mapping, as shown in Figure 1^[4]. The front-end abstracts camera data into a model that is easy to estimate, also known as VO. The back-end optimizes the model obtained by the front-end and obtains the globally consistent trajectory and map. The loop back detection mainly solves the cumulative error problem of VO over time, and mainly uses the word bag model. Mapping is the process of

constructing a map according to the estimated trajectory. The constructed map can be divided into metric map and topological map. Among them, metric map can be divided into sparse map and dense map.

Sensors are the means of perception of external environment in unmanned system, which are divided into endoplasmic and extrinsic. Endoplasmic sensors are used to feel the ontology information of unmanned aerial vehicle (UAV) system, mainly inertial measurement unit (IMU), laser and camera. The extrinsic sensors are mainly the GPS, which are installed in the environment and confined by the surroundings. In contrast, the camera, laser and other portable sensors are freer, cameras in VSLAM system are mainly divided into monocular cameras, binocular cameras (stereo cameras) Stereo, depth cameras (RGB-D) and other panoramic cameras (event camera), as shown in Figure 2.

Monocular	Binocular	RGB-D
•Low cost •Distance not limited •Scale uncertainty •Initialization problems	•Depth calculation •Distance not limited •Complex configuration •Large calculation	•Active testing •Good effect of reconstruction •Small range of measurement •Solar interference •Material interference

Figure 2. Comparison of three kinds of cameras in VSLAM.

B. Front-end (visual odometer)

A front-end is also called a visual odometer (VO), which estimates the motion between frames of the camera and the positions of the signposts. In VO, according to the image information obtained from the motion of the sensors, the orientation and trajectory of the corresponding sensors are estimated, in which the image information is transformed into a matrix and then the motion state of the sensor is obtained by solving the matrix. At present, the solutions of VO are mainly based on the feature point method of image feature changes and the photometric method (also called direct method) based on image gray level changes. The feature point method extracts the evenly distributed corner points and the surrounding descriptors from each frame image. Through the invariance of the descriptors around these corner points, the inter-frame matching is completed. Then, the camera poses are restored and the map coordinates are confirmed by epipolar geometry. Finally, the camera poses and map are fine-tuned according to the minimum reprojection error. The direct method determines the camera pose and map according to the photometric error, and does not need to extract corner points and descriptors. Because of this, the direct method cannot characterize the global features of an image. The elimination of cumulative drift in closed-loop detection of direct method has not been solved well. The advantages and disadvantages of the two methods are shown in Table 1.

Table 1. Comparison of advantages and disadvantages of the feature point method and direct method.

Methods	Feature point method	Direct method
Advantages	(a) No additional feature matching is required for loop back detection and repositioning. (b) Strong adaptability to changes in ambient light and camera exposure. (c) Interframe motion estimation with large scale can be realized.	(a) Not necessary to calculate descriptors and matching features, and the estimation efficiency of pose is higher under sparse condition. (b) Unsensitive to features and can work in areas where texture is not significant. (c) Free to build sparse, semi-sparse or dense maps.
Disadvantages	(a) Feature matching takes more time and the update rate of pose estimation is low. (b) Mismatched features need to be filtered. (c) Number of feature points is limited and sparse maps can only be constructed.	(a) A single pixel lacks distinction, and it is difficult to realize effective loopback detection. (b) Sensitive to scene change, only suitable for motion estimation between small-scale frames. (c) Sensitive to image photometric changes, camera exposure parameters need to be calibrated and limited in outdoor scenarios.

C. Back-end (filtering and nonlinear optimization)

The back-end accepts camera pose and environment information estimated by visual odometer and optimizes this information to obtain globally consistent motion trajectories and environment maps, which is based on optimizing trajectory and map state estimation problems from noisy data. There are two main ways for back-end optimization in VSLAM system: (i) filtering method represented by extended Kalman filter (EKF) and (ii) nonlinear optimization method represented by graph optimization.

According to the system state and motion input of the previous moment, the filtering method mainly uses Bayesian principle to estimate the state and confidence of the current moment. Then, optimal estimation of the current state is obtained based on the observation data and system state at the current time. The filter-based optimization method is relatively successful in small space, but its storage increases with the square of state, which indicating that its use in large unknown scenes is limited. The filtering methods include extended Kalman filter, unscented Kalman filter and particle filter. The core idea based on the nonlinear optimization (graph optimization) is to transform the back-end optimization algorithm into a form of graph, in which the vertices are set as the pose and environment characteristics at different time and the edges are set as the constraint relationship between vertices. After the map is built, the pose and environment characteristics are solved by the map optimization algorithm, so that the state to be optimized on the vertex can better meet the constraints of the corresponding edge. After the optimization algorithm is executed, the corresponding map is the motion trajectory and environment map.

D. Back loop detection

The task of loop back detection is to enable the unmanned system to identify the current scene based on sensor information and determine whether the area has been accessed when it returns to the original place, in which the error generated during the motion of the VSLAM system can be greatly eliminated, Figure 3^[5]. When it revisits a certain position, it should be able to correct or reduce these deviations and ensure the global consistency between the trajectory and the environmental map. The main method of loop back detection is to use the word bag model, which is to construct a word table containing k words by k mean clustering for the local features extracted from the image. According to the number of times each word appears in the word

table, the image is represented as a k dimension vector, which is used to judge the difference of the scene and to identify the accessed scene.

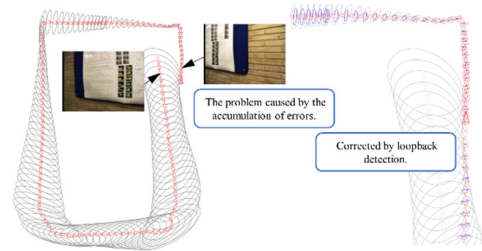


Figure 3. Cumulative error and loop detection correction results.

E. Mapping

According to the camera track and image, the map corresponding to the task requirement is established. Once the location of the road punctuation is determined, the map can be completed. In VSLAM system, the environment map refers to the set of all the road punctuation points obtained by the unmanned system in a period of time. Mapping is the process of the determination of all the road points in front-end detection and back-end optimization. VSLAM mapping mainly provides positioning services, and plays the role of navigation, obstacle avoidance and environmental reconstruction. When initializing or tracking missed, the unmanned system should be able to use the existing map information according to the current observed data and have the ability to recover from the wrong (or initial) location. Maps mainly include metric maps, topological maps, sparse maps and dense maps, as shown in Figure 4^[6].

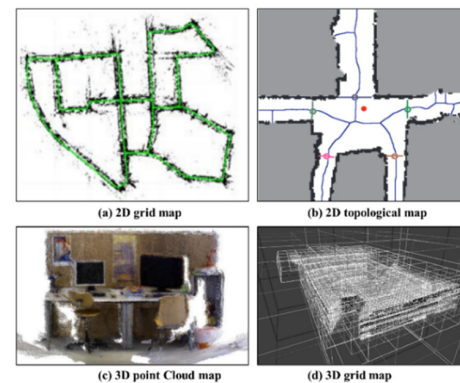


Figure 4. Different kinds of maps.

III. MATHEMATICAL DESCRIPTION AND ALGORITHM

The trajectory of unmanned system moving in the environment is described from x_1 to x_k . The road punctuation points in the environment are represented from y_1 to y_n . Then the equation for the motion of unmanned system from the moment $k-1$ to k can be expressed as follows,

$$x_k = f(x_{k-1}, u_k, w_k) \quad (1)$$

where u_k is the input information and w_k is the noise in the system. According to the observed road mark y_j at the moment k , the equation of the observed value $z_{k,j}$ is obtained, shown as follows,

$$z_{k,j} = h(y_j, x_k, v_{k,j}) \quad (2)$$

where x_k is the observation point and $v_{k,j}$ is the noise in the system.

As regards the research methods in VSLAM, the visual odometer (VO) mainly uses the feature point method (matching with feature points and building sparse maps) and direct method (building dense maps by minimizing photometric errors). The main problem of visual odometer (VO) is to estimate the motion of sensor by image information, in which it is necessary to select more representative points from the image and estimate the motion of the sensor by the position change of these points in the image. Since individual pixel values are easily changed by external conditions, artificial feature extractors are usually used to obtain robust image features, in which the features remain stable after the sensor moved. Researchers have designed many robust feature extraction algorithms. Davod Lowe et al. proposed the scale-invariant feature transformation (SIFT) algorithm and refined it in 2004, which is an image processing method used to detect and scan local features in images^[7]. MonoSLAM is the first real-time localization and map construction system successfully used in pure vision of monocular cameras proposed by Davison et al. and the heart of it is to create sparse but persistent natural landmarks (selecting meaningful parts) maps within the lines of the probabilistic framework. Mur-Artal et al. open the source of Oriented FAST and Rotated BRIEF (ORB)-SLAM algorithm, which is based on sparse features and plays a certain role in promoting the development of VSLAM. The same features are used for all SLAM tasks with ORB-SLAM algorithms: tracking, mapping, repositioning, and closed-loop, in which the selected reconstructed points and key frames are robust. The ORB-SLAM can generate compact traceable maps, which will grow only when the scene content changes and can be reused.

The direct method calculates the camera motion according to the pixel information of the image and it does not take a lot of time to calculate the feature points, in which, just by using feature points, the useful image information is considered. The direct method is divided into three types: sparse, dense and semi-dense, all of which have the ability to restore dense results. At present, many direct methods with ordinary camera as sensor have been proposed in VSLAM systems. A semi-direct visual odometry (SVO) algorithm was proposed, which is a sparse direct method. The SVO algorithm has the characteristics of precision, strong robustness and fast speed, and does not require

expensive feature extraction and robust matching techniques for motion estimation. Engel et al. proposed the direct sparse odometry (DSO) algorithm, which is a novel and high-precision VO measurement method based on sparse direct structure and motion formula. Kerl et al. proposed the DVO algorithm, which can quickly and accurately estimate the motion of RGB-D images. The depth camera trajectory is first estimated by image flow and then the cumulative error during dense VO modeling is eliminated by using a pose SLAM method based on key frame. The DVO method runs in real time on a single CPU kernel and it has a small and constant memory footprint, therefore, it is friendly for robots with limited computing resources.

The back-end algorithms in VSLAM systems are mainly divided into filtering-based and optimization-based algorithms. The filtering-based algorithm is to predict the state of moment k by moment $k-1$, while the optimization-based algorithm is to predict the complete trajectory and map of the robot by all the states before moment $k-1$. The method based on filtering originates from Bayesian filtering and is mainly divided into two parts: prediction and update. The prediction is based on the state of previous moment and input information to predict the state of the current moment, while the update is to match the currently observed sensor data with the map and correct the previously predicted state. These two steps iteratively and incrementally integrate sensor data to estimate robot pose and map. Since the filter-based method only considers the state of the previous moment and its incremental characteristics, it is also called online SLAM. Kalman filter (KF) is a linear optimal filter and two assumptions must be satisfied: (i) both the motion equation and the observation equation can be described by the linear equation, and (ii) all the states satisfy the Gaussian distribution^[8]. Therefore, Kalman filter can only be used in linear system. Most of the motion equations and observation equations of the system are nonlinear in real life. Moutarlier et al. extended KF to EKF, which is used to solve the probability estimation problem in nonlinear systems. The nonlinear filtering problem is transformed into an approximate linear filtering problem by Taylor series in EKF, but the EKF SLAM is sensitive to error data association. Therefore, an error observation will lead to a large error in filtering. In addition, because the complexity of EKF is twice the number of landmarks on the map, it is difficult to use it for large scale maps^[8].

Map optimization method was developed around 2004, which mainly uses Bundle Adjustment to solve the problem of pose change^[9]. It treats the pose and landmarks of the robot as vertices, and expresses the constraint relationship between them with the edges connected to them. Then, the Bundle Adjustment is applied to the system, and the pose estimation problem is transformed into the optimization problem, in which the constraint relationship of edges is better satisfied by adjusting the position of vertices in the map. Golfarelli et al. proposed the solution of the back end is equivalent to calculating the minimum energy of the model, in which the map optimization problem is regarded as a spring mass model^[10]. All observation and control information in robot motion is used to optimize the results of the whole trajectory and map construction, therefore, this method is also called Full SLAM. At last, the main algorithms for VSLAM are shown in Table 2.

Table 2. Development of main algorithms for VSLAM (ordered by Year)

Algorithms	Year	Sensor	Properties	Ref.
MonoSLAM	2007	Monocular	First real-time monocular VSLAM system.	[11]
PSLAM/PTAM	2007	Monocular	For the first time, nonlinear optimization and the key frame mechanism are used.	[12]
DTAM	2011	Monocular	Monocular dense map, direct method.	[13]
DVO	2013	RGBD	RGBD dense map, direct method.	[14]
RTAB-MAP	2013	Binocular/RGBD	Complete, high integration.	[15]
SVO	2014	Monocular	Sparse direct method.	[16]
LSD-SLAM	2014	Monocular	Successful application of monocular direct method, semi-dense map.	[17]
RGBD-SLAM-V2	2014	RGBD	Complete RGBD dense construction.	[18]
ORB-SLAM	2015	Mainly monocular	ORB, innovative use of three threads to complete SLAM.	[19]
Elastic Fusion	2015	RGBD	RGBD online reconstruction.	[20]
ROVIO	2015	Monocular+IMU	Based on the integration of EKF and IMU.	[21]
OKVIS	2015	Binocular+IMU	Tight coupling, based on nonlinear optimization.	[22]
DSO	2016	Monocular	Currently, the best monocular direct approach.	[23]
ORB-SLAM2	2016	Mainly monocular	ORB, improvement of the ORB-SLAM.	[23]
VINS	2017	Monocular	VIO based on optimization and sliding windows.	[24]

IV. RESEARCH SPOTS AND CHALLENGES

Conventional VSLAM methods still face the following problems: (i) the methods are sensitive to illumination and are not robust in environments with poor illumination conditions or complex illumination conditions; (ii) the tracking of feature point in traditional method is easy to lose when the camera motion amplitude is large; (iii) the processing for the dynamic object is not good enough; (iv) the amount of information to be calculated is large and the system response is slow. In order to solve the problems of traditional VSLAM, researchers have done a lot of innovative researches.

A. Combination of VSLAM and deep learning

The combination of deep learning and SLAM is a hot research direction in recent years. Deep learning is generally used in one or more parts of VSLAM systems, mainly for VO, closed-loop detection and semantic SLAM^[25]. A VSLAM system can collect and store images well, but it lacks depth analysis, recognition and mining of images. With deep learning, the VSLAM system can identify objects, which makes robots better understand the world and further expand from path planning to a wider range of task planning. The system can select different sensors and actuators according to different tasks. The combination of computer vision and deep learning has greatly improved the accuracy, execution efficiency and robustness of vision related tasks, such as instance classification, object detection, behavior recognition. Deep learning is based on high performance computers and GPU for big data and long-term training. The primary problem for the application of deep learning theory in VSLAM is how to provide sufficient computing power in embedded systems or advance the process of deep learning algorithms in small samples. VO estimate the camera motion by tracking the feature points between adjacent image frames and reconstruct the environment. The pose of the current frame is estimated by calculating the motion between frames in most cases of VO. The VO based on deep learning does not need complex geometric operations, and the end-to-end operation makes the method based on deep learning more concise.

The 3D maps of the surrounding environment and the position and direction of camera can be calculated in VSLAM.

The combination of deep learning and VSLAM holds the following four advantages: (i) VSLAM system based on deep learning has strong generalization ability and can work in complex light environment, in which the identification and processing for dynamic object are more effective; (ii) the model is trained by data-driven method, which is more in line with the law of the interaction between human and environment and has a lot of research and development space; (iii) the image information can be better fused with other sensor data by using neural network and the inter-frame drifts can be optimized, in which the extraction of semantic information is more efficiently (beneficial to the construction of semantic SLAM and the understanding of scene semantics); (iv) a series processes for front-end tracking and back-end optimization are eliminated in end-to-end VSLAM, and the estimated pose can be obtained by directly inputting the image. However, how to apply deep learning architecture throughout the SLAM system is still a great challenge.

B. Data processing and fusion in multi-sensor

With the development and popularization of various hardware sensors, the SLAM based on multi-sensor data fusion technology is also a key area and hot spot in academic and industrial research, in which researchers try to achieve the highest possible accuracy and robustness by utilizing the complementarity of various sensors. Multi-sensor data fusion is a kind of technology which combines several sensor sources to form a unified description, in order to improve the accuracy, increase the application range, reduce the complexity and improve the timeliness and robustness^[26]. The problems of map construction and global positioning of indoor robots in complex scenes such as shopping malls and corridors are solved by the combination of multi-sensor data processing and deep fusion and the SLAM algorithm based on map optimization, which has the advantages of real-time, good continuity, high robustness and high mapping and positioning accuracy.

It is very sensitive for data association in VSLAM, which can establish a relationship between multi-sensor data and other measured data to determine whether they have a common source. Data association is used to determine the relationship between measured data and features of map in VSLAM, in which the uncertainty of robot pose, the variation of feature density, the

interference of dynamic features in environment and the existence of observation errors make the processing of data association complex. The wrong association of data will not only cause the robot to deviate from the location, but also affect the created map and lead to the divergence of the algorithm. Ji et al. proposed a method for motion estimation and map creation by using the data from 3D laser scanner, image and IMU, in which an ordered multi-layer process from coarse-filtering to precise optimization is used^[27]. Firstly, the motion is predicted by the using of IMU, then the combination of IMU and vision is applied followed by lidar for frame-to-frame matching for further optimization and map creation. Therefore, the VSLAM can be used in high dynamic environments or complex environments with dark, no texture and no significant structures. The data fusion of multiple sensors can determine the corresponding relationships between the measured data and the feature sources of each sensor, and ensure that these corresponding relationships can play an optimal performance in complex environments. The deep neural network shows strong ability in matching multi-type data and processing data of different frequencies. VSLAM of multi-sensor fusion is an important development direction in the future.

C. SLAM for vision/inertial navigation

Driven by UAVs, unmanned vehicles, robots, reality enhancement, high-precision maps and other applications, the vision/inertial combined navigation technology has achieved rapid development. The projects, such as Tango and unmanned vehicles in Google, Hololens in Microsoft, ARKit in Apple, unmanned vehicles in Baidu, unmanned aerial vehicle in Dajinag, high-precision map in Gaode and other large-scale application projects, have set up visual/inertial combined navigation technology related researches, which has greatly promoted the application of them in the national economy^[28]. In addition to visual sensors, there are inertial and GPS navigation systems, in which errors can be accumulated in inertial devices and the acquisition is limited in GPS. To make up for the above defects, the real-time information provided by vision can be used for the fusion of inertia and GPS navigation, thus improving the navigation accuracy. The angular velocity and acceleration of the sensor itself can be measure directly in inertial measurement unit (IMU). However, the drift for the carrier of low speed is obvious and even a large amount of observation will be produced when it is still. This is the advantage of the camera, in which the motion structure in low speed can be restored. Visual-inertial odometry (VIO) is a kind of visual inertia combined navigation, and researchers have made some achievements in this field. Feng et al. deduced the observability condition of monocular vision/inertial combined navigation system based on high and low precision IMU, and proposed the filtering algorithm of both.

Although, in VSLAM, the requirements for computing resources are gradually decreasing and the related accuracy is gradually improving, the inherent defects such as slow data update (second level), significant decrease in measurement accuracy under high-speed motion, and difficulty in handling dynamic obstacles still cannot meet the strong maneuvering and fast response requirements of unmanned systems in dynamic environment. IMU has the advantages of high data update rate, good dynamic response, separable angular motion and linear motion. Since the IMU and vision sensors are complementary,

the inertial vision SLAM system can effectively improve the measurement bandwidth and anti-interference ability of the VSLAM system and enhance the robustness in the dynamic environment. It is the key direction of current VSLAM research. Sun et al. analyzed the basic principles and iconic achievements of visual/inertial combined navigation technology from the perspective of synchronous composition and positioning^[29]. Guo et al. comprehensively describe the visual/inertial combined navigation technology based on filter technology, especially the observability of filter and the consistency of filter state^[30]. The visual/inertial combined navigation technology has been widely used, but the reliability needs to be strengthened under complex conditions. Its future development is mainly reflected in the following three directions. The first is to improve the quality of information sources, including improving the zero-bias stability of inertial devices, environmental adaptability visual sensor illumination dynamic adaptability, fast motion adaptability and so on. The second is to improve the level of information fusion algorithm. Visual and inertial information have their own characteristics, and the quality of information is different under different conditions. It is necessary to design intelligent information fusion mechanism. And the third is the development of new navigation theory. Many animals in nature have amazing navigation skills, such as the Arctic terns can commute tens of thousands of kilometers from the south and north poles, and the carrier pigeons can return to the nest hundreds of kilometers away.

V. RESEARCH SPOTS AND CHALLENGES

A. Based on deep learning and deep estimation

There are still some problems in VSLAM based on traditional methods, such as sensitivity to illumination changes, unsatisfactory effect in dynamic scenes. Recently, deep learning theory has gained high attention in computer vision and other fields. Deep learning can help visual system to improve the accuracy and robustness of visual tasks to a great extent. There are two main directions for the application of deep learning to VSLAM, namely, replacing a module in VSLAM by neural network method and training semantic map with label by deep learning.

SLAM consists of many modules, such as closed-loop detection, image matching and pose estimation, in which better results can be achieved by the use of deep learning. Yi et al. proposed a new deep learning image feature LIFT, and uniformly completed the detection, direction estimation and feature description of it. Guo et al., CVPR meeting in 2019, proposed to construct the cost quantity to complete the binocular matching by grouping correlation, in which the left feature and the right feature are divided into two groups along the channel dimension and the correlation maps are calculated between each group to obtain multiple matching cost proposals. It is then packaged into cost quantities to complete stereo matching estimation of parallax between corrected image pairs. By adding semantic information, a map with labels can be obtained by using SLAM algorithm. The algorithm proposed by Tateno et al. can effectively fuse the semantic labels acquired from a single frame and dense SLAM, which can further produce semantic coherent scene reconstruction from a single view, as shown in

Figure 5^[31]. On the other hand, the label information is introduced into the objective functions and constraints of the Bundle Adjust (BA) or optimization end, which can be optimized by combining the feature point position and label information. Fioraio et al. proposed a semantic binding adjustment framework, which verified the hypothesis of verifying the existence and posture of the object sought for the global semantic optimization of the camera and object pose^[32].

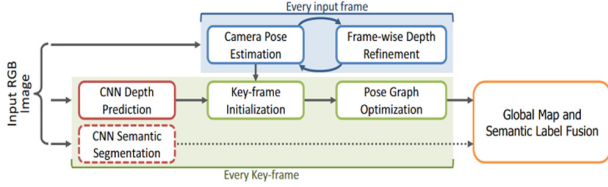


Figure 5. Flow for sematic coherent scene reconstruction from a single view.

The combination of VSLAM and deep learning is still at a preliminary stage. Convolutional neural networks in deep learning are constructed by simulating the way human brain nerves work. Although most researchers believe that convolutional neural network technology is a “black box” model, it does not hinder its great success in computer vision. The form of environment perception of VSLAM is carried out by image or video, so it is a very meaningful study to apply the method of deep learning to VSLAM. The VSLAM loop back detection algorithm combined with convolution neural network and Location Sensitive Hash (LSH) algorithm can significantly improve the detection efficiency and accuracy compared with the existing loop back detection algorithm. It can effectively reduce the influence of light, weather, angle and other changes on loopback detection, and meet the real-time requirements of VSLAM system. The investigation on how to build an end-to-end full VSLAM system based on deep learning is hoped in the future.

B. VSLAM in active and multiple robots

Conventional VSLAM mostly obtain the current position environment information from the data obtained by the camera, and further complete the map construction and location of the robots. For active VSLAM, in the unknown environment and without the human intervention, according to the information obtained, the robot can independently calculate the law of control, plan the motion path, locate the robot itself and construct the map. In the study of active VSLAM, mobile robots need to determine the motion law independently under the action of environment according to the explored environmental characteristics and obstacles. Frintrop et al. introduced active vision into VSLAM and achieved good results^[33]. Bosse et al. put forward a VSLAM method based on potential field for the environment with obstacles, but it is necessary to set up a number of guiding points in the environment to avoid the robot falling into the local minimum point trap^[34]. In order to solve the problem of map construction of new environment quickly and effectively, the active exploration of environment by robot and other unmanned systems is carried out in VSLAM.

In the field of robot, the coordination and cooperation of multi-robot is the urgent result of application and the inevitable

trend of technology development. Through the cooperation of multi-robots, some jobs that cannot be completed by single robots can be solved, or they can be used to simplify the work quantity and complexity of a single robot, thus improving the efficiency. A multi-robot VSLAM has two main directions: one is centralized, in which each robot builds a map of its current location and transmits local information to the central station where the inference is performed; the other is distributed, in which, without a central data center for information fusion, it uses local communication to acquire basically agreement. The multi-robot VSLAM algorithm is not as mature as the single-robot VSLAM algorithm at present, and it faces the problems of task planning, communication bandwidth, distance limitation, map fusion and so on.

C. Semantic VSALM

Most VSLAM systems are metric maps or topological maps based on simple environmental geometric attributes, and such maps waste storage space and lack environmental details, which makes it difficult for unmanned systems to achieve more advanced environmental perception (like human beings) in addition to navigation and positioning. How to complete the upgrade from path planning to task planning while improving the accuracy and robustness of autonomous positioning in VSLAM, and how to achieve more advanced human-computer interaction and environmental perception are the hot spots in this area. Based on the previous analysis, semantic VSLAM emerges as the times require. The related research results of semantic VSLAM can be divided into two categories: VSLAM auxiliary semantics and semantic auxiliary VSLA. The former is also called semantic mapping, which is mostly based on Convolutional Neural Networks (CNNs). Tateno et al. studied the method of monocular camera deep estimation based on CNNs and used stochastic gradient descent method to segment scene target semantically^[35]. John et al. developed a pixel-level density map and 2D semantic map based on CNNs, and then constructed the Semantic Fusion of 3D semantic maps to effectively enhance the ability of target recognition in dynamic environment^[36]. Ma et al. studied the RGB-D semantic mapping method based on FuseNet-CNN, in which, by applying the multi-view continuous test optimization method, the detection uncertainty under single view was greatly alleviated^[37].

Compared with the semantic mapping in full swing, the related research of semantic auxiliary VSLAM is still in its infancy. Javier et al. first tried the monocular semantic SLAM method based on objective recognition optimization^[38]. Sean et al., based on the Expectation Maximization (EM) algorithm, studied the data association method between semantic information and feature point location, and then classified semantic information into re-cast error function^[39]. It is the most representative achievement in semantic auxiliary VSLAM at present. Then, on the previous basis, The semantic information as a mid-term constraint was proposed, further reducing the cumulative translation error in VSLAM long-term localization. Chao et al. combined the scene semantic segmentation results with motion consistency detection to improve the robustness of VSLAM in dynamic environments^[40].

VI. CONCLUSION

As an important part of autonomous navigation and perception of unmanned systems, VSLAM technology is currently in a period of vigorous development. The feature-based front-end inter-frame pose estimation is still the mainstream of the current VSLAM. The emergence of DSO, LSD-SLAM and other algorithms also marks the beginning of the direct method from theory to practice. The semi-direct method represented by SVO, which combines the advantages of the two parts has also attracted wide attention. Although the factor map optimization algorithm cannot shake the dominant position of pose map in global optimization of back-end, with the popularization of related algorithms and open-source code, there will be more mature open-source SLAM systems based on factor map optimization in the future. The tight coupling VI-SLAM has become an indispensable part of the VSLAM because of the limited characteristics of the visual sensor itself; it is foreseeable that the open-source SLAM framework will further develop the multi camera and VI versions after the release of the original monocular pure vision version.

Despite decades of research, VSLAM have matured in pose estimation, map construction, precision and real-time. However, there are still many challenges in dynamic adaptability, generalization ability, advanced sensing ability and multi-sensor integration, which have given birth to hot research directions such as event cameras, deep learning, semantic VSLAM. As one of the most attractive research directions in the field of artificial intelligence, the good self-organization and self-learning ability of deep learning make it shine brilliantly in feature extraction, loop-back detection. Although the research on semantic VSLAM is not perfect, as one of the most potential research directions, more mature research results and open-source framework will be available in the future. Currently, for VSLAM related applications, the mainly future directions of development are lightweight and miniaturization, so that VSLAM can run well on small devices such as embedded or mobile phones, and then the applications based on it can be considered. On the other hand, the precise 3D reconstruction, perfect reconstruction of scenes and scene understanding can be achieved by the utilization of high-performance computing equipment. Moreover, the VSLAM based on line/plane features, VSLAM in dynamic scenarios and multi-robot VSLAM are the hot directions of VSLAM related researches. The development of VSLAM has gone through three stages: Issues raised, Search Algorithm and Improvement of Algorithm. And, now, we are in the third stage, in the face of how to further improve the existing framework, so that the VSLAM system can operate stably under various interference conditions, which requires the unremitting efforts of researchers.

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